

Exercício Comentado sobre Semântica Denotacional

Considere a semântica denotacional do seguinte programa

$$y := 1; \text{ while } x \neq 1 \text{ do } \{y := y * x; x := x - 1\}$$

Aplice a função resultante a um estado s_0 tal que $s_0 x = 3$, e indique o valor da variável y no estado de chegada.

Sugere-se que siga os seguintes passos:

1. Construa $\mathcal{S}_{\text{ds}}\llbracket y := 1; \text{ while } x \neq 1 \text{ do } \{y := y * x; x := x - 1\} \rrbracket$ identificando a funcional F envolvida.
2. Calcule as várias funções $F^n \perp$ usadas na definição de $\text{FIX } F$ e apresente uma definição de $\text{FIX } F$.
3. Tem agora todos os dado para calcular o valor de y no estado de chegada.

Resolução

1.

$$\begin{aligned} & \mathcal{S}_{\text{ds}}\llbracket y := 1; \text{ while } x \neq 1 \text{ do } \{y := y * x; x := x - 1\} \rrbracket s_0 \\ &= (\mathcal{S}_{\text{ds}}\llbracket \text{while } x \neq 1 \text{ do } \{y := y * x; x := x - 1\} \rrbracket \circ \mathcal{S}_{\text{ds}}\llbracket y := 1 \rrbracket) s_0 \\ &= \mathcal{S}_{\text{ds}}\llbracket \text{while } x \neq 1 \text{ do } \{y := y * x; x := x - 1\} \rrbracket (s_0[y \mapsto 1]) \\ &= (\text{FIX } F)(s_0[y \mapsto 1]) \end{aligned}$$

onde $(F g) = \text{cond}(\mathcal{B}\llbracket x \neq 1 \rrbracket, g \circ \mathcal{S}_{\text{ds}}\llbracket y := y * x; x := x - 1 \rrbracket, \text{id})$
ou seja,

$$\begin{aligned} (F g) s &= \begin{cases} g(\mathcal{S}_{\text{ds}}\llbracket y := y * x; x := x - 1 \rrbracket s) & \text{se } \mathcal{B}\llbracket x \neq 1 \rrbracket s = \mathbf{tt} \\ \text{id } s & \text{se } \mathcal{B}\llbracket x \neq 1 \rrbracket s = \mathbf{ff} \end{cases} \\ &= \begin{cases} g((\mathcal{S}_{\text{ds}}\llbracket x := x - 1 \rrbracket \circ \mathcal{S}_{\text{ds}}\llbracket y := y * x \rrbracket) s) & \text{se } s x \neq 1 \\ s & \text{se } s x = 1 \end{cases} \\ &= \begin{cases} g((\mathcal{S}_{\text{ds}}\llbracket x := x - 1 \rrbracket (\mathcal{S}_{\text{ds}}\llbracket y := y * x \rrbracket s)) & \text{se } s x \neq 1 \\ s & \text{se } s x = 1 \end{cases} \\ &= \begin{cases} g(\mathcal{S}_{\text{ds}}\llbracket x := x - 1 \rrbracket (s[y \mapsto (s y) \times (s x)])) & \text{se } s x \neq 1 \\ s & \text{se } s x = 1 \end{cases} \\ &= \begin{cases} g(s[y \mapsto (s y) \times (s x)][x \mapsto (s x) - 1]) & \text{se } s x \neq 1 \\ s & \text{se } s x = 1 \end{cases} \end{aligned}$$

2. Relembre que $\perp$ é a função que, para todo o estado s , $\perp s = \underline{\text{undef}}$, e que

$$(F\ g)\ s = \begin{cases} g(s[y \mapsto (s\ y) \times (s\ x)][x \mapsto (s\ x) - 1]) & \text{se } s\ x \neq 1 \\ s & \text{se } s\ x = 1 \end{cases}$$

Para simplificar a apresentação, vamos assumir que $s' = s[y \mapsto (s\ y) \times (s\ x)][x \mapsto (s\ x) - 1]$

$$(F^0\ \perp)\ s = \perp\ s = \underline{\text{undef}}$$

$$(F^1\ \perp)\ s = \begin{cases} \perp\ s' & \text{se } s\ x \neq 1 \\ s & \text{se } s\ x = 1 \end{cases} = \begin{cases} \underline{\text{undef}} & \text{se } s\ x \neq 1 \\ s & \text{se } s\ x = 1 \end{cases}$$

$$(F^2\ \perp)\ s = (F(F\ \perp))\ s = \begin{cases} (F\ \perp)\ s' & \text{se } s\ x \neq 1 \\ s & \text{se } s\ x = 1 \end{cases} = \begin{cases} \underline{\text{undef}} & \text{se } s\ x \neq 1 \wedge s'\ x \neq 1 \\ s' & \text{se } s\ x \neq 1 \wedge s'\ x = 1 \\ s & \text{se } s\ x = 1 \end{cases}$$

$$= \begin{cases} \underline{\text{undef}} & \text{se } s\ x \neq 1 \wedge (s\ x) - 1 \neq 1 \\ s[y \mapsto (s\ y) \times (s\ x)][x \mapsto (s\ x) - 1] & \text{se } s\ x \neq 1 \wedge (s\ x) - 1 = 1 \\ s & \text{se } s\ x = 1 \end{cases}$$

$$= \begin{cases} \underline{\text{undef}} & \text{se } s\ x \neq 1 \wedge s\ x \neq 2 \\ s[y \mapsto (s\ y) \times 2][x \mapsto 1] & \text{se } s\ x = 2 \\ s & \text{se } s\ x = 1 \end{cases}$$

$$(F^3\ \perp)\ s = (F(F^2\ \perp))\ s = \begin{cases} (F^2\ \perp)\ s' & \text{se } s\ x \neq 1 \\ s & \text{se } s\ x = 1 \end{cases}$$

$$= \begin{cases} \underline{\text{undef}} & \text{se } s\ x \neq 1 \wedge s'\ x \neq 1 \wedge s'\ x \neq 2 \\ s'[y \mapsto (s'\ y) \times 2][x \mapsto 1] & \text{se } s\ x \neq 1 \wedge s'\ x = 2 \\ s' & \text{se } s\ x \neq 1 \wedge s'\ x = 1 \\ s & \text{se } s\ x = 1 \end{cases}$$

$$= \begin{cases} \underline{\text{undef}} & \text{se } s\ x \neq 1 \wedge (s\ x) - 1 \neq 1 \wedge (s\ x) - 1 \neq 2 \\ s'[y \mapsto (s'\ y) \times 2][x \mapsto 1] & \text{se } s\ x \neq 1 \wedge (s\ x) - 1 = 2 \\ s' & \text{se } s\ x \neq 1 \wedge (s\ x) - 1 = 1 \\ s & \text{se } s\ x = 1 \end{cases}$$

$$= \begin{cases} \underline{\text{undef}} & \text{se } s\ x \neq 1 \wedge s\ x \neq 2 \wedge s\ x \neq 3 \\ s[y \mapsto (s\ y) \times (s\ x) \times 2][x \mapsto 1] & \text{se } s\ x = 3 \\ s[y \mapsto (s\ y) \times (s\ x)][x \mapsto (s\ x) - 1] & \text{se } s\ x = 2 \\ s & \text{se } s\ x = 1 \end{cases}$$

$$= \begin{cases} \underline{\text{undef}} & \text{se } s\ x \neq 1 \wedge s\ x \neq 2 \wedge s\ x \neq 3 \\ s[y \mapsto (s\ y) \times 3 \times 2][x \mapsto 1] & \text{se } s\ x = 3 \\ s[y \mapsto (s\ y) \times 2][x \mapsto 1] & \text{se } s\ x = 2 \\ s & \text{se } s\ x = 1 \end{cases}$$

$\vdots$

$$(F^n\ \perp)\ s = \begin{cases} \underline{\text{undef}} & \text{se } s\ x < 1 \vee s\ x > n \\ s[y \mapsto (s\ y) \times (s\ x) \times \dots \times 1][x \mapsto 1] & \text{se } 1 \leq (s\ x) \leq n \end{cases}$$

Portanto,

$$(\text{FIX } F) s = \begin{cases} \underline{\text{undef}} & \text{se } s x < 1 \\ s[y \mapsto (s y) \times (s x) \times \dots \times 1][x \mapsto 1] & \text{se } s x \geq 1 \end{cases}$$

3. O estado s_0 é tal que $s_0 x = 3$. Portanto, o valor da variável y no estado de chegada é

$$\begin{aligned} (\mathcal{S}_{\text{ds}}[y := 1; \text{while } x \neq 1 \text{ do } \{y := y * x; x := x - 1\}] s_0) y &= ((\text{FIX } F)(s_0[y \mapsto 1])) y \\ &= (s_0[y \mapsto 1 \times 3 \times 2 \times 1][x \mapsto 1]) y \\ &= 1 \times 3 \times 2 \times 1 \\ &= 6 \end{aligned}$$